

ELKHORN SLOUGH RESERVE, CA, USA

WMO: 997997

Lat: 36.815N

Lon: 121.738W

Elev: 10

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	35.3	37.5	23.0	17.3	49.4	27.0	21.0	47.8	18.8	55.2	16.5	55.2	2.1	50	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	13.2	77.6	62.1	73.9	60.9	70.6	61.4	66.4	70.6	65.2	69.4	63.8	68.2	5.7	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	93.3	67.7	63.7	88.3	66.6	62.1	83.5	65.1	31.0	69.7	30.0	69.4	29.0	68.2	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	13.3	11.5		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.9	53.2	51.6	52.9	53.3	55.3	57.7	59.4	60.3	60.6	59.9	55.2	51.1
	DBStd	5.45	5.74	5.37	4.26	3.89	3.28	2.99	2.81	3.30	3.81	4.74	5.14	4.79
	HDD50	157	35	39	17	12	0	0	0	0	0	0	10	44
	HDD65	3361	367	375	376	351	303	222	175	151	142	171	296	432
	CDD50	2311	134	85	107	112	164	231	291	318	316	308	167	78
	CDD65	41	0	1	1	1	2	3	2	4	9	14	4	0
	CDH74	303	6	7	13	9	19	18	8	12	61	120	30	0
	CDH80	54	0	0	3	0	2	7	1	0	11	28	2	0
(14) Wind	WSAvg	4.8	4.6	4.7	4.9	5.9	6.0	5.6	5.3	4.8	4.3	4.0	3.7	4.0
(15) Precipitation	PrecAvg	17.5	3.8	3.6	2.5	1.3	0.6	0.1	0.0	0.0	0.0	0.9	1.6	3.4
	PrecMax	28.9	9.6	12.8	9.3	3.7	2.0	0.6	0.0	0.3	0.4	4.0	5.2	7.8
	PrecMin	9.2	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
	PrecStd	4.8	2.8	2.8	2.4	1.1	0.6	0.2	0.0	0.1	0.1	1.1	1.4	2.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.7	74.7	74.2	74.7	76.8	78.9	72.1	76.4	80.5	83.0	78.6	68.0
		MCWB	55.9	58.3	60.0	60.5	60.5	61.2	65.3	65.0	64.3	60.8	59.5	57.4
	2%	DB	70.0	68.3	66.6	67.3	67.5	69.2	69.2	71.4	74.4	77.0	73.0	64.3
		MCWB	56.8	55.9	57.8	58.7	57.5	61.6	64.1	65.0	62.9	60.9	58.0	56.4
	5%	DB	66.3	63.1	63.5	63.1	63.9	65.8	67.1	69.0	70.2	71.7	67.1	61.3
		MCWB	54.3	55.4	56.8	56.5	57.7	60.5	63.2	63.7	62.9	61.3	55.8	55.4
	10%	DB	62.0	59.8	60.5	60.5	61.5	63.8	64.9	66.6	67.6	68.1	63.5	58.9
		MCWB	53.3	53.9	55.2	54.9	56.7	59.3	61.3	62.2	62.0	60.4	55.0	54.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.4	61.3	62.7	62.8	62.1	65.8	67.8	68.3	66.4	68.0	62.1	62.2
		MCDB	67.5	67.7	70.5	72.4	70.7	68.7	71.1	73.7	73.2	72.6	71.5	65.2
	2%	WB	58.2	59.0	60.0	59.4	59.9	63.3	65.7	66.2	65.2	65.5	59.9	60.1
		MCDB	63.3	64.3	64.7	65.0	65.4	67.5	67.6	69.8	71.6	70.1	66.5	62.1
	5%	WB	56.5	56.9	57.9	57.3	58.4	61.2	63.9	64.4	63.8	63.7	58.4	58.2
		MCDB	62.4	61.9	61.8	61.7	63.0	65.1	66.5	67.9	69.2	68.3	63.6	59.5
	10%	WB	55.0	54.8	56.0	55.5	57.2	59.4	61.8	62.5	62.4	61.8	56.8	56.1
		MCDB	60.8	58.9	59.7	58.9	61.2	63.0	64.5	66.0	67.1	66.3	61.5	57.5
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	16.2	16.1	14.9	13.0	11.7	9.9	10.5	13.2	17.1	17.7	15.9
		MCDBR	22.8	21.3	21.1	20.7	18.2	16.2	13.1	14.9	18.8	23.9	23.5	17.8
		MCWBR	12.6	12.0	12.9	13.0	11.1	9.8	8.1	8.8	10.8	12.6	11.9	10.8
	5% WB	MCDBR	18.0	17.4	18.8	17.7	15.2	14.3	12.2	14.4	16.8	17.5	19.7	12.3
		MCWBR	11.2	11.0	12.3	11.7	9.9	8.8	8.3	8.9	10.0	11.3	11.2	9.2
(40) Clear-Sky Solar Irradiance	taub		0.320	0.325	0.326	0.352	0.363	0.354	0.357	0.341	0.350	0.323	0.311	0.315
	taud		2.546	2.539	2.536	2.438	2.422	2.458	2.470	2.503	2.471	2.579	2.580	2.575
	Ebn at Noon		278	290	299	295	291	293	291	294	287	287	279	272
	Edh at Noon		25	28	30	36	37	36	35	33	32	26	23	23
(44) All-Sky Solar Radiation	RadAvg		820	1101	1551	1972	2284	2492	2349	2121	1809	1350	950	723
	RadStd		93	122	129	166	132	97	68	77	65	79	51	92

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)		+0.72	N/A	-2.16	N/A	N/A	N/A	N/A	-216	+263	+65

Nomenclature: See separate page